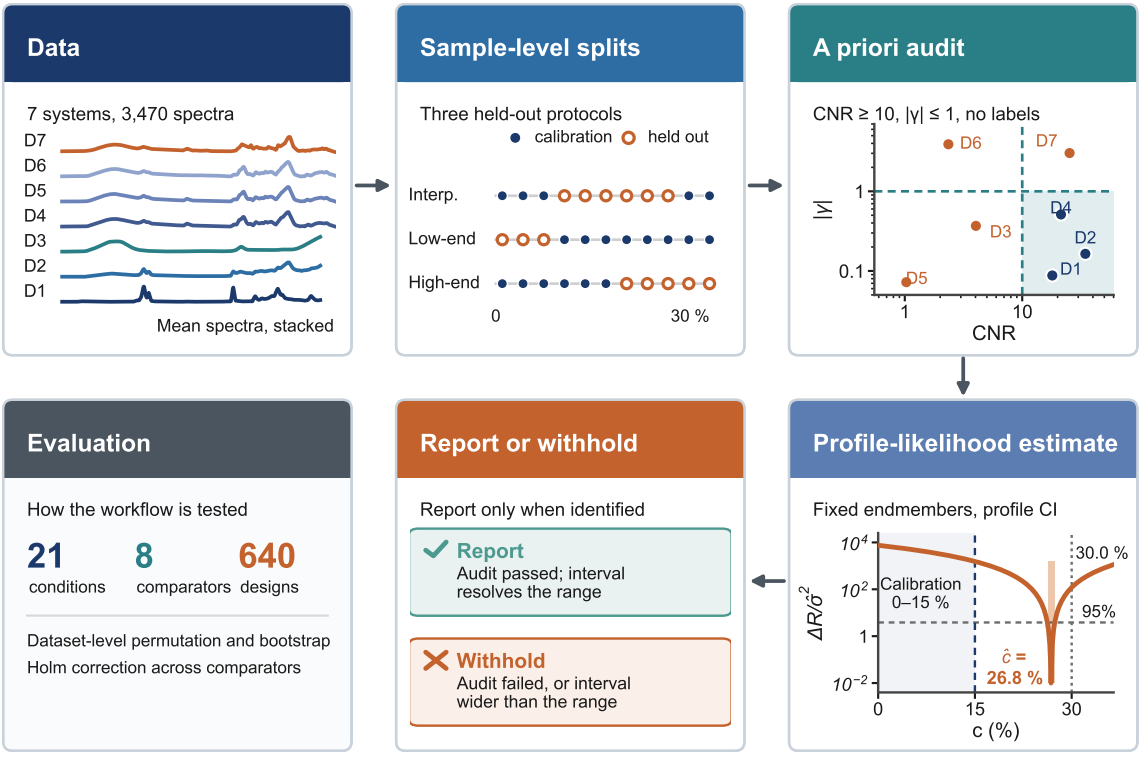

Data
7 systems, 3,470 spectra
D7
D6
D5
D4
D3
D2
D1
Mean spectra, stacked
Sample-level splits
Three held-out protocols
calibration
held out
Interp.
Low-end
High-end
0
30 %
A priori audit
CNR ≥ 10, |γ| ≤ 1, no labels
D6
D7
D4
D3
D2
D1
D5
|γ|
1
0.1
1
10
CNR
Profile-likelihood estimate
Fixed endmembers, profile CI
10^4
10^2
1
10^{-2}
$\Delta R/\hat{\sigma}^2$
Calibration
0–15 %
30.0 %
95%
$\hat{c}$ =
26.8 %
0
15
30
c (%)
Report or withhold
Report only when identified
Report
Audit passed; interval
resolves the range
Withhold
Audit failed, or interval
wider than the range
Evaluation
How the workflow is tested
21
conditions
8
comparators
640
designs
Dataset-level permutation and bootstrap
Holm correction across comparators